# Query-Oriented Dynamic Multimodal Alignment for Few-Shot Hyperspectral Image Classification

Qiaoli Zhang, Jiangtao Peng, *Senior Member, IEEE*, and Weiwei Sun, *Senior Member, IEEE*

***Abstract*—Deep learning (DL) has advanced hyperspectral image classification (HSIC), but label scarcity remains a significant challenge. Traditional unimodal methods usually produce unstable prototypes, while multimodal methods suffer from cross-modal semantic misalignment. Moreover, standard metrics fail to capture high-order feature correlations, further limiting the discriminative ability of the model. To address these issues, we propose a query-oriented dynamic multimodal alignment (QODMA) method, which integrates visual-textual guidance with dual-distance metric learning (DML) for few-shot HSIC. Specifically, a query-oriented dynamic attention (QODA) module is designed to bridge the modality gap by aligning visual and textual features through query-driven attention interactions. A bidirectional attention mechanism is constructed to employ contrastive learning to enhance intraclass compactness. Additionally, a DML module that combines the Euclidean distance (ED) and Brownian distance covariance (BDC) metrics is employed to refine the feature space representation, thereby enhancing the discriminative ability of the model. To mitigate domain shift, an inter-domain structural consistency loss (IDSCL) is constructed. Experimental results on four public hyperspectral datasets demonstrate that the proposed QODMA outperforms state-of-the-art few-shot classification methods.**

***Index Terms*—Few-shot learning (FSL), hyperspectral image classification (HSIC), multimodality, prototypical network (PNet).**

## I. Introduction

HYPERSPECTRAL images (HSIs) capture a wide range of spectral information, which is crucial for land cover classification [1], [2]. Traditional HSI classification (HSIC) methods usually relied on manually designed spectral features [3], [4], which may struggle in complex environments. Recently, deep learning (DL) has significantly advanced HSIC. Chen et al. [5] first used the stacked autoencoder (SAE) approach to learn spatial-spectral features. Due to the large number of parameters in SAE, they further proposed a convolutional neural network (CNN)-based method for HSIC [6]. Subsequently, many CNN variants have been proposed [7]. However, as the network depth increases, CNN-based methods may encounter gradient issues. Zhong et al. [8] proposed a spectral-spatial residual network (SSRN) to solve these problems. Although these HSIC methods have shown promising results, they usually require a large number of labeled samples. For real applications, we usually face the small sample classification problem due to the high cost of labeling HSI. It is difficult to train a good DL model with limited labeled samples [9], [10], [11].

To address the challenge of limited labeled samples, few-shot learning (FSL) has emerged as an effective solution [12], [13], [14]. FSL aims to learn effective prior knowledge from a small number of labeled samples, which is then used to make predictions on unlabeled samples. Recently, many FSL methods based on embedding and metric learning have been proposed for HSIC [15]. Typically, these methods learn a transformation function by constructing meta-tasks to map samples into an embedding space. Subsequently, distance-based metric methods are used for classification. Snell et al. [16] proposed a prototypical network (PNet), which learns the prototype of each class (i.e., the mean of sample features in that class) and uses the Euclidean distance (ED) for classification. Then, Liu et al. [17] introduced the PNet into HSIC and proposed a deep FSL (DFSL) framework. However, the labeled samples from the target domain were not utilized during the training phase, leading to the loss of prior knowledge from the target domain. To overcome this drawback, a series of cross-domain FSL methods for HSIC have been proposed [18], [19], [20], [21]. Li et al. [22] proposed a deep cross-domain FSL (DCFSL) method for HSIC, which mitigates domain shift through conditional adversarial domain adaptation. Liu et al. [19] proposed a refined prototypical contrastive learning (RPCL) method for HSIC, which combines triple constraints to stabilize prototypes and introduces a fusion training strategy to reduce domain shift. Li et al. [23] proposed a spectral coordinate transformer (SCFormer) for cross-domain HSIC, which incorporates dense spectral coordinate blocks within the encoder to enhance the learning of generalized feature representations. Zhu et al. [24] proposed a cycle-resemblance few-shot transformation (CF-Trans) network for cross-domain HSIC, which introduces a cyclic-resemblance adversarial method to capture domain invariant knowledge, thereby improving the generalization performance of the model.

Although the above methods have demonstrated good classification performance, they all use the ED in the latent space.

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It has a potential assumption that different feature dimensions have uniform variance. For high-dimensional data, features are often correlated and have different variances. To address the variance anisotropy problem, several studies have introduced second-order moment-based metrics, such as covariance matrices [25] or Gaussian models that combine first-order and second-order moments [26]. Xi et al. [27] proposed a class covariance metric-based FSL method for HSIC. Sun et al. [28] proposed a few-shot HSIC method based on the Earth mover's distance (EMD) metric.

Additionally, most current FSL methods heavily rely on the visual modality, which struggles to capture semantic attributes from very few samples, resulting in weak semantic representation in image prototypes. Inspired by the human ability to extract information from multiple modalities, such as vision and language, recent research has incorporated multimodal information into few-shot classification to enhance semantic representations and improve classification performance [29], [30], [31]. Radford et al. [32] proposed a contrastive language-image pretraining (CLIP) model, which utilizes contrastive learning to integrate information from visual and linguistic modalities. Inspired by the CLIP, Li et al. [33] proposed a knowledge-guided semantic transfer network, which utilizes a category-guided visual learning module and a knowledge transfer network to propagate category correlations. Li et al. [34] proposed a cross-domain few-shot HSIC framework with cross-modal alignment and supervised contrastive learning (CDFS-CASCL), incorporating semantic prior knowledge from labels. However, they did not consider the semantic differences between modalities. Specifically, the visual modality usually focuses on spatial and spectral features, while the textual modality emphasizes semantic aspects. The semantic differences between modalities will affect the effectiveness of contrastive learning.

To address these challenges, we propose a query-oriented dynamic multimodal alignment (QODMA) framework for few-shot HSIC, which contains three modules: a query-oriented dynamic attention (QODA) module, a dual-distance metric learning (DML) module, and an inter-domain structural consistency loss (IDSCL) module. In the QODA module, bidirectional attention scores between visual and textual features are computed through query exchange. A gating mechanism dynamically adjusts cross-modal feature consistency by weighing each modality's contribution according to its relevance to the query, ensuring that only the most salient information is emphasized. In the DML module, the ED and Brownian distance covariance (BDC) are combined to capture both local similarities and global higher-order dependencies. Finally, the IDSCL module aligns the source and target domains. In summary, the main contributions of this article are as follows.

1) A QODMA framework is proposed for few-shot HSIC, which integrates visual and linguistic information from HSIs, effectively transfers meta-knowledge to the target domain through three modules, i.e., QODA, DML, and IDSCL.
2) A QODA module is designed to bridge the modality gap by dynamically aligning visual and textual features through query-driven attention interactions. The module alleviates the modality difference through bidirectional attention, enhancing the feature consistency between two modalities.
3) A DML module is constructed to integrate the ED and the BDC, which captures local similarities as well as global high-order feature dependencies to enhance the discriminative ability of the model.

The rest of this article is organized as follows. Section II describes the proposed QODMA methodology. The experimental evaluations and analysis are provided in Section III. Finally, Section IV concludes this article.

## II. Proposed Method

### A. Data Processing

Assume the source domain dataset $D_s$ contains $C_s$ classes, while the target domain dataset $D_t$ contains $C_t$ classes, where $C_s \geq C_t$ and $D_s \cap D_t = \emptyset$. In FSL, all samples in $D_s$ are labeled, while only a limited number of labeled samples are available in $D_t$ (e.g., five labeled samples per class in this article). Gaussian noise augmentation is used to expand the number of labeled samples in $D_t$ from 5 to 200/class.

Both the source and target domains follow the same episodic training mode under the $N$-way $K$-shot setting. For each meta-task, $N$ categories are randomly selected, and then $K$ samples from the support set and $K_q$ samples from the query set are randomly selected from each category. Taking the source domain as an example, the support and query sets are represented as $S_s = \{(\boldsymbol{x}_i^{\mathrm{ss}}, y_i^{\mathrm{ss}})\}_{i=1}^{N\times K}$, $Q_s = \{(\boldsymbol{x}_j^{\mathrm{sq}}, y_j^{\mathrm{sq}})\}_{j=1}^{N\times K_q}$, respectively. In this article, the parameters are set as $N = C_t$, $K = 1$, and $K_q = 19$.

### B. Overview

Fig. 1 illustrates the framework of the proposed QODMA, which comprises three main modules, i.e., QODA module, DML module, and IDSCL module.

During the meta-training process, an adaptive mapping layer is employed to align the spectral dimensions between the source and target domains. The visual and textual encoders then extract rich features from HSIs and texts, respectively. To align visual and text features, a QODA module consisting of a query-driven bidirectional attention mechanism with contrastive learning is constructed. In the FSL metric learning stage, a DML module is designed to measure both local similarities and high-order statistical dependencies by combining the ED with the BDC metric. In addition, the IDSCL module is used to align the source and target features. During the meta-testing process, a small number of labeled target samples are used to train a nearest neighbor classifier to classify the rest unlabeled samples.

### C. Visual and Textual Feature Extraction

A dual-branch visual encoder is employed to extract both spectral and spatial features from HSIs, as illustrated in Fig. 2. The network consists of a series of convolutional and residual operations. The spectral branch uses $1 \times 1$ convolutions to model channel-wise spectral dependencies. This point-wise

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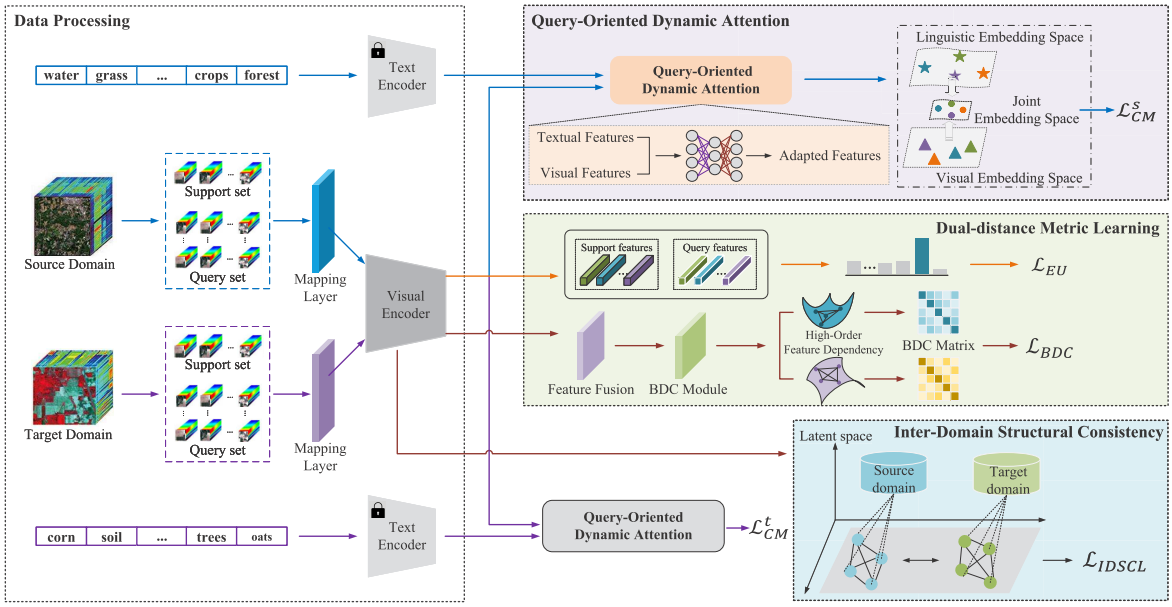



Fig. 1. Framework of the proposed QODMA. It consists of three modules, i.e., QODA, DML, and IDSCL. The QODA module adopts query-driven attention interaction to dynamically align visual and textual features. The DML module combines the Euclidean metric and the BDC metric to improve the discriminative ability. The IDSCL module mitigates the domain gap between source and target domains.

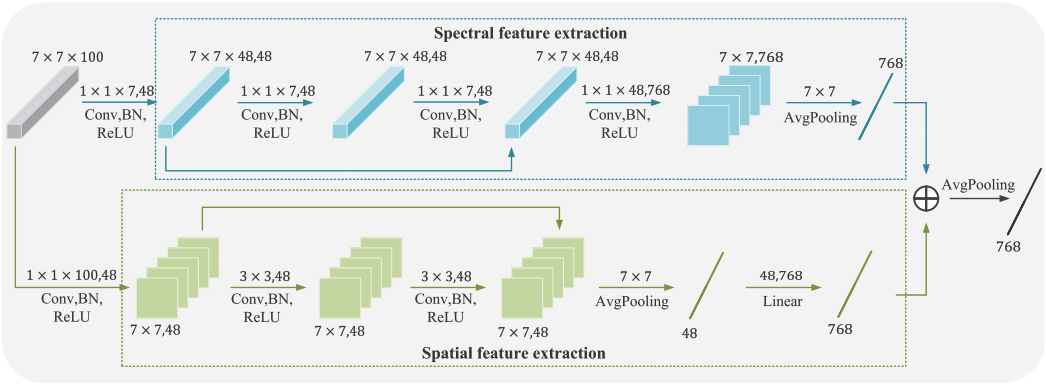



Fig. 2. Flowchart of visual encoder.

operation preserves unique spectral signatures of each pixel. In contrast, the spatial branch adopts 3 × 3 convolutions to capture local spatial textures. The 3 × 3 kernel strikes a balance between capturing meaningful spatial information and computational efficiency. The outputs from both branches are fused to produce the final visual spectral and spatial features.

For textual feature extraction, each class label is encoded using the Base BERT model [35]. The input label is first tokenized into a sequence with two special markers: [CLS] at the beginning and [SEP] at the end. The token sequence is then processed by the 12-layer BERT encoder, where each layer employs multihead self-attention, feed-forward networks, and layer normalization to produce contextualized token embeddings. The final hidden state corresponding to the [CLS] token is extracted as the aggregated semantic representation of the input label, resulting in a 768-D feature vector that captures label-level semantic information suitable for subsequent integration with visual features.

### D. Dual-Distance Metric Learning

To learn discriminative features, two complementary distance metrics, i.e., the ED and BDC, are used to capture both local similarity and high-order statistical dependencies.

For both source and target domains, each episode corresponds to a meta-task, which includes a support set and a query set. Taking the source domain as an example, the visual prototype of the $c$-ch category in the support set is computed as

$$\boldsymbol{P}_c^s = \frac{1}{K} \sum_{\boldsymbol{x}_i^{\mathrm{ss}} \in S_s^c} f_\theta\left(\boldsymbol{x}_i^{\mathrm{ss}}\right), \quad c = 1, \ldots, N \tag{1}$$

where $f_\theta(.)$ represents the mapping layer and visual encoder.

The probability of a query sample $\boldsymbol{x}_j^{\mathrm{sq}} \in Q_s$ belonging to the category $c_j$ is calculated as

$$p\left(y_j = c_j | \boldsymbol{x}_j^{\mathrm{sq}}\right) = \frac{\exp\left(-\mathrm{dist}\left(f_\theta\left(\boldsymbol{x}_j^{\mathrm{sq}}\right), \boldsymbol{P}_{c_j}^s\right)\right)}{\sum_{c_j=1}^{N} \exp\left(-\mathrm{dist}\left(f_\theta\left(\boldsymbol{x}_j^{\mathrm{sq}}\right), \boldsymbol{P}_{c_j}^s\right)\right)} \tag{2}$$

where dist(·) represents the ED.

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The FSL loss function based on the ED metric for the source domain is represented as

$$\mathcal{L}_{\mathrm{EU}}^{s} = \mathbb{E}_{S_s,Q_s}\left[-\sum_{\left(\boldsymbol{x}_j^{\mathrm{sq}},y_j\right)\in Q_s}\log p\left(y_j = c_j|\boldsymbol{x}_j^{\mathrm{sq}}\right)\right] \tag{3}$$

where $\mathbb{E}_{(S_s,Q_s)}$ denotes the expectation of the loss over all episodes generated via the episodic sampling process in FSL.

Similarly, the $\mathcal{L}_{\mathrm{EU}}^{t}$ for the target domain follows the same procedure. Finally, the FSL loss for both domains is

$$\mathcal{L}_{\mathrm{EU}} = \mathcal{L}_{\mathrm{EU}}^{s} + \mathcal{L}_{\mathrm{EU}}^{t}. \tag{4}$$

To further capture high-order feature dependencies, the BDC metric is introduced. It quantifies the difference between the joint and marginal feature distributions and has a closed-form solution for discrete observations [36], [37], [38]. Here, we use a lightweight BDC module that contains two convolutional layers: one for dimensionality reduction and the other for BDC matrix computation.

Specifically, assume that the network output feature map is $\mathbf{Z} = f'_\theta(\boldsymbol{x}_i^{\mathrm{ss}}) \in \mathbb{R}^{h\times w\times d}$, which is reshaped it into $\mathbf{Z} \in \mathbb{R}^{hw\times d}$. Each column of the reshaped matrix is considered as a random vector of observations. Next, we compute an initial ED matrix $\tilde{\mathbf{G}}$ that captures the pairwise distances between the columns of $\mathbf{Z}$

$$\tilde{\mathbf{G}} = \sqrt{2\left(\mathbf{1}\left(\mathbf{Z}^T\mathbf{Z}\odot\mathbf{I}\right)_{\mathrm{sym}} - 2\mathbf{Z}^T\mathbf{Z}\right.} \tag{5}$$

where $\mathbf{1}$ is a matrix with all entries equal to 1, and $\mathbf{I}$ is the identity matrix. $(\cdot)_{\mathrm{sym}}$ indicates the symmetric part of the matrix. The symbol $\odot$ denotes the Hadamard product. Subsequently, the BDC matrix $\mathbf{G}$ is computed from $\tilde{\mathbf{G}}$ as

$$\mathbf{G} = \sqrt{\tilde{\mathbf{G}}} - \frac{2}{d}\left(\mathbf{1}\sqrt{\tilde{\mathbf{G}}}\right)_{\mathrm{sym}} + \frac{1}{d^2}\mathbf{1}\sqrt{\tilde{\mathbf{G}}}\mathbf{1}. \tag{6}$$

We denote the operation that computes the BDC matrix from an input feature map as $\mathbf{G}_\theta(\cdot)$. Taking the source domain as an example, the spatial and spectral features before the average pooling layer are first fused and dimensionally reduced, and then fed into the BDC module to obtain the BDC matrix corresponding to the $c$th prototype

$$\tilde{\boldsymbol{P}}_c^s = \frac{1}{K}\sum_{\boldsymbol{x}_i^{\mathrm{ss}}\in S_s^c}\mathbf{G}_\theta\left(f'_\theta\left(\boldsymbol{x}_i^{\mathrm{ss}}\right)\right), \quad c = 1,\ldots,N \tag{7}$$

where $f'_\theta(.)$ represents the fused feature map output.

The probability of a query sample $\boldsymbol{x}_j^{\mathrm{sq}} \in Q_s$ belonging to category $c_j$ is calculated as

$$\tilde{p}\left(y_j = c_j|\boldsymbol{x}_j^{\mathrm{sq}}\right) = \frac{\exp\left(\mathrm{tr}\left(\mathbf{G}_\theta\left(f'_\theta\left(\boldsymbol{x}_j^{\mathrm{sq}}\right)\right)^T\tilde{\boldsymbol{P}}_{c_j}^s\right)\right)}{\sum_{c_j=1}^{N}\exp\left(\mathrm{tr}\left(\mathbf{G}_\theta\left(f'_\theta\left(\boldsymbol{x}_j^{\mathrm{sq}}\right)\right)^T\tilde{\boldsymbol{P}}_{c_j}^s\right)\right)} \tag{8}$$

where $\mathrm{tr}(\cdot)$ denotes the matrix trace. The loss based on the BDC metric is computed as follows:

$$\mathcal{L}_{\mathrm{BDC}}^{s} = \mathbb{E}_{S_s,Q_s}\left[-\sum_{\left(\boldsymbol{x}_j^{\mathrm{sq}},y_j\right)\in Q_s}\log\tilde{p}\left(y_j = c_j|\boldsymbol{x}_j^{\mathrm{sq}}\right)\right]. \tag{9}$$

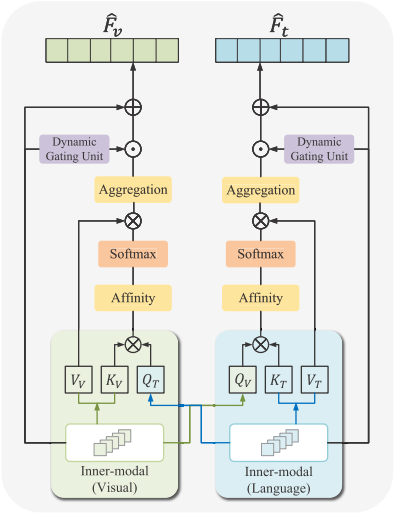



Fig. 3. Flowchart of the QODA module.

Similarly, we can compute the BDC loss for the target domain and obtain the overall BDC loss as

$$\mathcal{L}_{\mathrm{BDC}} = \mathcal{L}_{\mathrm{BDC}}^{s} + \mathcal{L}_{\mathrm{BDC}}^{t}. \tag{10}$$

Finally, the overall FSL loss for each episode, combining both the ED loss and BDC loss, is given by

$$\mathcal{L}_{\mathrm{FSL}} = \mathcal{L}_{\mathrm{EU}} + \mathcal{L}_{\mathrm{BDC}}. \tag{11}$$

### *E. Query-Oriented Dynamic Attention*

CLIP-based few-shot image classification methods usually focus on the alignment between modalities, while ignoring the inherent semantic inconsistency between modalities. To address this issue, we propose a QODA module as shown in Fig. 3. QODA computes a bidirectional attention score between visual and textual features via query exchange. Then, a gating mechanism dynamically weighs the contribution of each modality based on its relevance to the query, ensuring that only the most relevant information is emphasized. This process mitigates the semantic differences between modalities, thereby learning more consistent and representative cross-modal features.

The first phase implements attention-driven adaptive refinement through bidirectional query-key-value attention between modalities. Specifically, we design a dual-path cross-attention mechanism where the query is dynamically exchanged between modalities while keys and values remain anchored to their original modality.

Taking the source domain as an example, given a visual features $f_\theta(\boldsymbol{x}_i^{\mathrm{ss}}) \in \mathbb{R}^{d_v}$ and its corresponding textual features $f_\varphi(\boldsymbol{t}_i^{\mathrm{ss}}) \in \mathbb{R}^{d_t}$. For textual-to-visual attention

$$\mathbf{F}_{v\leftarrow t}^{s} = \mathrm{softmax}\left(\frac{f_\varphi\left(\boldsymbol{t}_i^{\mathrm{ss}}\right)\mathbf{W}_Q'^{\top}\cdot f_\theta\left(\boldsymbol{x}_i^{\mathrm{ss}}\right)\mathbf{W}_K^{\top}}{\sqrt{d}}\right)\cdot f_\theta\left(\boldsymbol{x}_i^{\mathrm{ss}}\right)\mathbf{W}_V^{\top}. \tag{12}$$

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For visual-to-textual attention

$$\mathbf{F}^s_{t\leftarrow v} = \text{softmax}\left(\frac{f_\theta\left(\boldsymbol{x}_i^{\text{ss}}\right)\mathbf{W}_Q^\top \cdot f_\varphi\left(\boldsymbol{t}_i^{\text{ss}}\right)\mathbf{W}_K'^\top}{\sqrt{d}}\right)\cdot f_\varphi\left(\boldsymbol{t}_i^{\text{ss}}\right)\mathbf{W}_V'^\top \tag{13}$$

where $\mathbf{W}_Q$, $\mathbf{W}_K$, $\mathbf{W}_V$, and $\mathbf{W}'_Q$, $\mathbf{W}'_K$, $\mathbf{W}'_V$ are modality-specific projection weights. The exchanged queries allow each modality to actively "interrogate" the other, creating two complementary interaction paths.

To refine the bidirectional attention outputs, we introduce a learnable gating block that dynamically adjusts the cross-modal feature consistency. This modulation mechanism adaptively balances the contributions of visual and textual modalities based on their relevance to the query, ensuring effective feature fusion and reducing semantic misalignment

$$\hat{\mathbf{F}}^s_t = f_\varphi\left(\boldsymbol{t}_i^{\text{ss}}\right)\mathbf{W}'_Q + \gamma_t \odot \mathbf{F}^s_{t\leftarrow v} \tag{14}$$

where $\odot$ denotes the Hadamard product. The gating weights $\gamma_v, \gamma_t \in [0, 1]$.

After cross-modal interaction via bidirectional attention and dynamic gating, we enforce feature alignment through contrastive learning. For the $i$th paired sample in a batch of $N$ pairs, the contrastive loss is

$$\ell^s_{\text{cl}}\left(\hat{\mathbf{F}}^s_{v_i}, \hat{\mathbf{F}}^s_{t_i}\right) = -\log\left(\frac{\exp\left(-\frac{\text{dist}\left(\hat{\mathbf{F}}^s_{v_i}, \hat{\mathbf{F}}^s_{t_i}\right)}{\tau}\right)}{\sum_{k=1,k\neq i}^{2N}\exp\left(-\frac{\text{dist}\left(\hat{\mathbf{F}}^s_{v_i}, \hat{\mathbf{F}}^s_{t_k}\right)}{\tau}\right)}\right) \tag{15}$$

where dist$(\cdot)$ denotes the similarity measurement function based on the ED, and $\tau$ is the temperature coefficient ($\tau = 0.5$). The total cross-modal loss combines both directional perspectives

$$\mathcal{L}^s_{\text{CM}} = \frac{1}{2N}\sum_{i=1}^{N}\left(\ell^s_{\text{cl}}\left(\hat{\mathbf{F}}^s_{v_i}, \hat{\mathbf{F}}^s_{t_i}\right) + \ell^s_{\text{cl}}\left(\hat{\mathbf{F}}^s_{t_i}, \hat{\mathbf{F}}^s_{v_i}\right)\right). \tag{16}$$

Similarly, the cross-modal loss for the target domain follows the same procedure. Thus, the overall cross-modal loss for both the source and target domains is

$$\mathcal{L}_{\text{CM}} = \mathcal{L}^s_{\text{CM}} + \mathcal{L}^t_{\text{CM}}. \tag{17}$$

### F. Inter-Domain Structural Consistency Loss

To reduce the domain shift, an IDSCL is introduced to maintain a consistent autocorrelation structure between source and target domain feature maps. Specifically, we first extract intermediate feature maps from the visual encoder of the source and target domains

$$m_s = f'_\theta\left(\boldsymbol{x}^{\text{ss}}\right), \quad m_t = f'_\theta\left(\boldsymbol{x}^{\text{ts}}\right) \tag{18}$$

where $m_s \in \mathbb{R}^{\text{hw}\times d}$ and $m_t \in \mathbb{R}^{\text{hw}\times d}$ represent the feature matrices obtained by flattening the spatial dimensions of the extracted feature maps.

To model the intrinsic feature structure, we compute the auto-correlation matrices for the source and target domain feature maps

$$M_s = \frac{m_s^T m_s}{\|m_s\|_F}, \quad M_t = \frac{m_t^T m_t}{\|m_t\|_F} \tag{19}$$

where $\|\cdot\|_F$ denotes the Frobenius norm.

Then, the IDSCL is defined as

$$\mathcal{L}_{\text{IDSCL}} = \frac{1}{N^2}\sum_{i=1}^{N}\sum_{j=1}^{N}\|M_s^{(i)} - M_t^{(j)}\|_F \tag{20}$$

where $M_s^{(i)}$ and $M_t^{(j)}$ denote the auto-correlation matrices of the $i$th and $j$th class prototypes in the source and target domains, respectively.

The total loss for the model can be expressed as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{FSL}} + \mathcal{L}_{\text{CM}} + \mathcal{L}_{\text{IDSCL}}. \tag{21}$$

**Algorithm 1** QODMA Training Process

**Input**:
The visual feature and textual feature extractors $f_\theta$, $f_\varphi$
The label text sets of source and target domains $T_s$, $T_t$
The support set $S_s$ and query set $Q_s$ in $D_s$
The support set $S_t$ and query set $Q_t$ in $D_t$
Number of training episodes $E$
**Output:** The loss of model

```
if episode < E do
    Train with S_s, Q_s, S_t, Q_t, T_s, and T_t
    Calculate prototypes by Eq. (1) and Eq. (7)
for x_i^ss ∈ S_s, x_j^sq ∈ Q_s do
    Calculate the L_EU^s by Eq. (3)
    Calculate the L_BDC^s by Eq. (9)
end for
Calculate the L_EU^t same as L_EU^s
Calculate the L_BDC^t same as L_BDC^s
Calculate the L_FSL by Eq. (11)
for x_i^ss ∈ S_s, t_i^ss ∈ T_s do
    Calculate the L_CM^s by Eq. (16)
end for
Calculate the L_CM^t same as L_CM^s
Calculate the L_CM by Eq. (17)
for x_i^ss ∈ S_s, x_i^ts ∈ S_t do
    Calculate the L_IDSCL by Eq. (20)
end for
    Return the L_total of an episode by Eq. (21)
end if
```

The detail of the training procedure of the QODMA method is shown in Algorithm 1.

## III. Experimental Results and Analysis

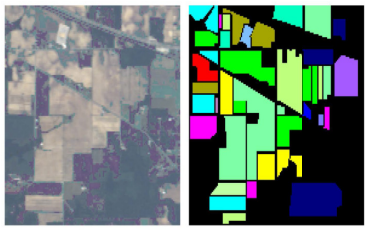
### A. Datasets

In the experiments, we adopt the Chikusei (CH) dataset as the source domain $D_s$ and use the Indian Pines (IP), University of Pavia (UP), Salinas (SA), and Houston (HT) hyperspectral datasets as the target domain $D_t$, respectively. Table I lists the

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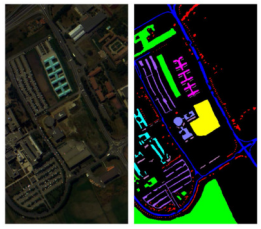
TABLE I
DATASET DETAILS FOR FOUR TARGET DOMAIN DATASETS: IP, UP, SA, AND HT

**Indian Pines (IP)**



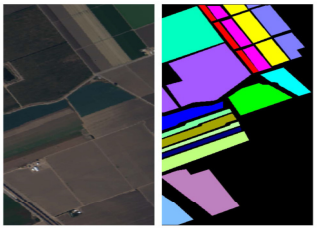
| Class | Land-covers | Pixels | Color |
|---|---|---|---|
| 1 | Alfalfa | 46 | |
| 2 | Corn-notill | 1428 | |
| 3 | Corn-mintill | 830 | |
| 4 | Corn | 237 | |
| 5 | Grass-pasture | 483 | |
| 6 | Grass-trees | 730 | |
| 7 | Grass-pasture-mowed | 28 | |
| 8 | Hay-windrowed | 478 | |
| 9 | Oats | 20 | |
| 10 | Soybean-notill | 972 | |
| 11 | Soybean-mintill | 2455 | |
| 12 | Soybean-clean | 593 | |
| 13 | Wheat | 205 | |
| 14 | Woods | 1265 | |
| 15 | Buildings-Grass-Trees-Drives | 386 | |
| 16 | Stone-Steel-Towers | 93 | |
| | Total | 10249 | |

**University of Pavia (UP)**



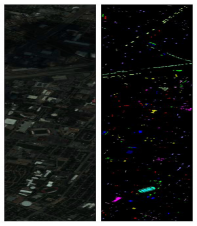
| Class | Land-covers | Pixels | Color |
|---|---|---|---|
| 1 | Asphalt | 6631 | |
| 2 | Meadows | 18649 | |
| 3 | Gravel | 2099 | |
| 4 | Trees | 3064 | |
| 5 | Sheets | 1345 | |
| 6 | Bare soil | 5029 | |
| 7 | Bitumen | 1330 | |
| 8 | Bricks | 3682 | |
| 9 | Shadow | 947 | |
| | Total | 42776 | |

**Salinas (SA)**



| Class | Land-covers | Pixels | Color |
|---|---|---|---|
| 1 | Brocoli_green_weeds_1 | 2009 | |
| 2 | Brocoli_green_weeds_2 | 3726 | |
| 3 | Fallow | 1976 | |
| 4 | Fallow_rough_plow | 1394 | |
| 5 | Fallow_smooth | 2678 | |
| 6 | Stubble | 3959 | |
| 7 | Celery | 3579 | |
| 8 | Grapes_untrained | 11271 | |
| 9 | Soil_vinyard_develop | 6203 | |
| 10 | Corn_senesced_green_weeds | 3278 | |
| 11 | Lettuce_romaine_4wk | 1068 | |
| 12 | Lettuce_romaine_5wk | 1927 | |
| 13 | Lettuce_romaine_6wk | 916 | |
| 14 | Lettuce_romaine_7wk | 1070 | |
| 15 | Vinyard_untrained | 7268 | |
| 16 | Vinyard_vertical_trellis | 1807 | |
| | Total | 54129 | |

**Houston (HT)**



| Class | Land-covers | Pixels | Color |
|---|---|---|---|
| 1 | Healthy grass | 1251 | |
| 2 | Stressed grass | 1254 | |
| 3 | Synthetic grass | 697 | |
| 4 | Trees | 1244 | |
| 5 | Soil | 1242 | |
| 6 | Water | 325 | |
| 7 | Residential | 1268 | |
| 8 | Residential | 1244 | |
| 9 | Road | 1252 | |
| 10 | Highway | 1227 | |
| 11 | Railway | 1235 | |
| 12 | Parking Lot 1 | 1233 | |
| 13 | Parking Lot 2 | 469 | |
| 14 | Tennis Court | 428 | |
| 15 | Running Track | 660 | |
| | Total | 15029 | |

corresponding ground truth maps, land cover classes, sample numbers, and corresponding color labels for the four target domain datasets.

*1) Source Domain Dataset:* Chikusei (CH) was captured using the Headwall Hyperspec-VNIR-C sensor in Chikusei. It contains 128 spectral bands covering the wavelength range of 343–1018 nm, with a spatial resolution of 2.5 m. The image size is 2517 $\times$ 2335, and the dataset includes 19 categories.

*2) Target Domain Dataset:*

*a) Indian Pines:* The IP dataset was captured in 1992 by the airborne visible/infrared imaging spectrometer (AVIRIS) over a stand of IP trees in Indiana, USA. It consists of 200 spectral bands, covering wavelengths 400–2500 nm, and has an image size of 145 $\times$ 145 pixels. The dataset contains 16 categories and 10 249 labeled samples.

*b) University of Pavia:* The UP dataset was captured in 2003 by the reflective optics spectrographic imaging system (ROSIS-03) over the city of Pavia, Italy. The original data includes 115 spectral bands spanning a wavelength range of 0.43–0.86 $\mu$m, with a spatial resolution of 1.3 m. After removing 12 noisy bands, 103 spectral bands are retained for classification tasks. The image size is 610 $\times$ 340, with nine land cover classes and 42 776 labeled samples.

*c) Salinas:* The SA dataset was captured by the AVIRIS over the Salinas Valley, CA, USA. It offers a spatial resolution of 3.7 m and initially includes 224 spectral bands, of which 204 are retained after removing noisy bands. The image size is 512 $\times$ 217 pixels, with 16 categories and 54 129 labeled samples.

*d) Houston:* The HT dataset was introduced by the 2013 GRSS data fusion competition. The dataset includes 349 $\times$ 1905 pixels at a 2.5 m spatial resolution, comprising 144 spectral bands spanning 364–1046 nm, and contains 15 categories.

## B. Experimental Setting

To evaluate the performance of the proposed QODMA, we compare it with several classical HSIC methods, such as HybridSN [7] and SSRN [8], and six latest cross-domain FSL methods, including DCFSL [22], Gia-CFSL [18], RPCL-FSL [19], CPSRCL [39], ADAFSL [21], and CDFS-CASCL [34].

For all methods, only five labeled samples per class from the target domain are used. For HybridSN and SSRN, these samples are used to train the model for direct classification of the remaining unlabeled target samples. For the FSL methods, the experimental setup follows their original papers. Specifically, for DCFSL, Gia-CFSL, RPCL-FSL, CPSRCL, ADAFSL, CDFS-CASCL, and our method, 200 labeled samples per class are used from the source domain, while five labeled samples per class are used from the target domain.

We employ an episodic training strategy to construct $N$-way $K$-shot classification tasks on the support set, where $N$ denotes the number of test classes, and $K$ is set to 1. Each class in the query set includes 19 labeled samples.

All experiments are implemented using the PyTorch framework on a system with an Intel Xeon CPU, 64 GB RAM, and an NVIDIA GeForce RTX 2080 Ti GPU with 12 GB memory. The trainable parameters are optimized using the SGD method, with a learning rate set to 0.001. To evaluate classification performance, we adopt three standard metrics: overall accuracy (OA), average accuracy (AA), and $\kappa$ coefficient. Each experiments is run ten times, the averaged results are reported.

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TABLE II
CLASSIFICATION RESULTS (%) ON IP

| Class | HybridSN [7] | SSRN [8] | DCFSL [22] | Gia-CFSL [18] | RPCL-FSL [19] | CPSRCL [39] | ADAFSL [21] | CDFS-CASCL [34] | QODMA |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 51.75 | 81.65 | 91.46 | 95.58 | 98.79 | 93.91 | **99.56** | 98.78 | 99.19 |
| 2 | 53.59 | 63.91 | 40.23 | 51.85 | 74.07 | 66.48 | 59.51 | **75.52** | 72.25 |
| 3 | 44.68 | 42.54 | 59.21 | 55.84 | 56.48 | 73.09 | 65.24 | 79.33 | **79.60** |
| 4 | 43.76 | 40.93 | 85.31 | 79.74 | 96.55 | 83.62 | 85.49 | **98.49** | 97.70 |
| 5 | 65.53 | **87.91** | 78.03 | 69.10 | 82.01 | 67.36 | 84.19 | 66.95 | 81.24 |
| 6 | 87.02 | 93.82 | 85.59 | 85.98 | 93.79 | 89.59 | 96.43 | 97.31 | **97.66** |
| 7 | 31.62 | 47.79 | **100.00** | 97.60 | **100.00** | **100.00** | **100.00** | **100.00** | **100.00** |
| 8 | 97.13 | 95.38 | 91.75 | 84.15 | 98.45 | 79.28 | 90.88 | 83.72 | **99.65** |
| 9 | 15.35 | 27.76 | **100.00** | **100.00** | **100.00** | **100.00** | **100.00** | **100.00** | **100.00** |
| 10 | 57.82 | 51.12 | 66.70 | 62.01 | 80.14 | 78.96 | 65.67 | **83.09** | 73.70 |
| 11 | 75.89 | 76.06 | 49.69 | 53.92 | 57.12 | 62.29 | 70.64 | 68.94 | **79.01** |
| 12 | 63.97 | 39.53 | 43.28 | 54.10 | 75.34 | 68.45 | 75.61 | 90.81 | **91.72** |
| 13 | 84.31 | 85.31 | 97.50 | 97.92 | 98.05 | 99.17 | 96.83 | 99.50 | **99.67** |
| 14 | **95.53** | 93.18 | 87.22 | 89.04 | 92.06 | 85.32 | 82.57 | 88.98 | 88.89 |
| 15 | 63.63 | 71.97 | 60.24 | 73.13 | 69.29 | 92.39 | 82.76 | **96.77** | 93.44 |
| 16 | 65.98 | 60.86 | 93.75 | 97.56 | **100.00** | **100.00** | 94.32 | 98.86 | **100.00** |
| OA | 64.22 ±2.71 | 66.16 ±2.56 | 63.77 ±2.32 | 66.25 ±2.35 | 75.67 ±2.71 | 74.45 ±1.78 | 70.55 ±2.17 | 80.67 ±1.63 | **83.72 ±0.69** |
| AA | 62.15 ±2.53 | 66.48 ±4.63 | 76.86 ±0.03 | 77.97 ±1.61 | 86.05 ±1.21 | 83.79 ±1.56 | 79.10 ±1.04 | 89.05 ±1.07 | **90.87 ±0.46** |
| $\kappa$ | 60.24 ±2.05 | 62.43 ±2.69 | 59.41 ±2.18 | 62.13 ±2.81 | 72.78 ±2.84 | 71.25 ±1.57 | 65.73 ±2.14 | 78.33 ±1.79 | **81.58 ±0.75** |

TABLE III
CLASSIFICATION RESULTS (%) ON UP

| Class | HybridSN [7] | SSRN [8] | DCFSL [22] | Gia-CFSL [18] | RPCL-FSL [19] | CPSRCL [39] | ADAFSL [21] | CDFS-CASCL [34] | QODMA |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 67.01 | 90.83 | 65.44 | 72.56 | 89.79 | 89.94 | 87.86 | 91.34 | **91.76** |
| 2 | 80.52 | **89.08** | 78.59 | 82.18 | 74.56 | 82.74 | 78.42 | 76.89 | 83.25 |
| 3 | 44.45 | 56.11 | 62.06 | 62.32 | **96.34** | 65.92 | 75.60 | 91.29 | 87.18 |
| 4 | 35.75 | 89.15 | 91.76 | 92.81 | **98.91** | 94.46 | 90.59 | 74.27 | 88.79 |
| 5 | 91.73 | 97.97 | 99.14 | 98.43 | 90.49 | 99.33 | 96.72 | 99.87 | **99.93** |
| 6 | 74.04 | 53.46 | 76.34 | 71.58 | 90.04 | 89.78 | 87.88 | 73.41 | **90.16** |
| 7 | 73.51 | 49.19 | 76.91 | 80.63 | 93.13 | 85.24 | 62.66 | 92.13 | **98.69** |
| 8 | 61.90 | 71.13 | 69.81 | 65.91 | 90.13 | **92.35** | 79.39 | 87.02 | 87.64 |
| 9 | 76.42 | 96.50 | 98.99 | 98.85 | 99.68 | 98.46 | **99.79** | 99.68 | 98.25 |
| OA | 72.96 ±2.73 | 76.78 ±2.99 | 78.66 ±2.69 | 80.41 ±0.84 | 83.86 ±2.18 | 83.69 ±2.02 | 81.62 ±3.38 | 81.81 ±2.93 | **87.68 ±1.09** |
| AA | 67.25 ±2.85 | 77.29 ±3.06 | 80.58 ± 1.29 | 80.51 ±1.42 | 85.19 ±1.01 | 84.73 ±0.81 | 84.98 ±1.50 | 87.22 ±0.95 | **91.07 ±1.76** |
| $\kappa$ | 62.73 ±3.17 | 70.28 ±3.72 | 72.54 ±3.05 | 74.83 ± 0.67 | 79.18 ±2.71 | 78.67 ±2.32 | 76.52 ±3.54 | 76.78 ±2.94 | **83.64 ±1.09** |

### C. Comparison With Other Methods

Tables II–V show the classification results of various methods on the four datasets, respectively. The best results in each group are highlighted in bold.

Table II presents the classification results on the IP dataset. It is clear that our QODMA method consistently outperforms other methods in terms of OA, AA, and $\kappa$. Compared with the traditional DL method, SSRN and the latest cross-domain FSL method CDFS-CASCL, our method outperforms the OA by 17.56% and 3.05%, respectively. Our method attains the highest performance in 9 of the 16 ground-truth classes. Specifically, Classes 11 "Soybean-mintill" and 12 "Soybean-clean" have similar spectral properties, which makes them prone to confusion. Our QODMA shows the best results on these classes, outperforming other algorithms by at least 2.95% and 0.91%, respectively. This improvement may be partially attributed to the introduction of semantic information through the QODA module, which enhances the model's ability to distinguish spectrally similar classes by aligning visual features with class-level textual semantics.

Table III presents the classification results on the UP dataset. Our QODMA provides the highest OA, AA, and $\kappa$ coefficient with values of 87.68%, 91.07%, and 83.64%, respectively. Compared with the latest FSL-based methods, e.g., CDFS-CASCL and RPCL-FSL, the QODMA method improves the OA by 5.87% and 3.82%, respectively. In the UP dataset, Classes 1 "Asphalt" and 7 "Bitumen" share similar spectral properties, making them difficult to classify. Nevertheless, the QODMA still achieves the best classification performance on these classes.

Table IV presents the classification results on the SA dataset. Our QODMA still obtains the highest OA, AA, and $\kappa$ with values of 94.29%, 96.92%, and 93.66%, respectively. Among

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TABLE IV
CLASSIFICATION RESULTS (%) ON SA

| Class | HybridSN [7] | SSRN [8] | DCFSL [22] | Gia-CFSL [18] | RPCL-FSL [19] | CPSRCL [39] | ADAFSL [21] | CDFS-CASCL [34] | QODMA |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 98.20 | **100.00** | 99.03 | 99.13 | 99.05 | 98.95 | 99.63 | 99.78 | 99.70 |
| 2 | 91.54 | 94.34 | 99.61 | 99.69 | 99.52 | 100.00 | 99.82 | **100.00** | **100.00** |
| 3 | 77.14 | 87.29 | 86.86 | 92.67 | 90.61 | 91.50 | **95.82** | 94.14 | 94.88 |
| 4 | 78.82 | 82.58 | **99.89** | 99.53 | 98.69 | 99.86 | 97.60 | 99.28 | 99.50 |
| 5 | 98.64 | 94.05 | 90.53 | 86.74 | 89.81 | 90.97 | 96.66 | 95.90 | **99.18** |
| 6 | 78.23 | 98.26 | 99.24 | 98.68 | 97.95 | 99.95 | 99.41 | **100.00** | **100.00** |
| 7 | 93.87 | 98.94 | **100.00** | 98.95 | 99.05 | 99.87 | 99.14 | 99.80 | 99.86 |
| 8 | 83.66 | 76.04 | 81.60 | 82.79 | 77.32 | 81.55 | 82.33 | **87.31** | 83.22 |
| 9 | 83.14 | 99.49 | 99.71 | 94.14 | 99.71 | 98.97 | 99.55 | 99.24 | **99.94** |
| 10 | 85.96 | 89.99 | 67.87 | 80.25 | 86.43 | 86.86 | 88.94 | 87.52 | **92.00** |
| 11 | 84.99 | 95.06 | 98.31 | 96.99 | 99.32 | 99.11 | 98.71 | 99.53 | **100.00** |
| 12 | 92.60 | 97.52 | 99.91 | **99.95** | 98.77 | 99.69 | 98.80 | 99.82 | 99.90 |
| 13 | 87.64 | 97.91 | 99.01 | 99.12 | 99.46 | 99.12 | 99.71 | 99.95 | **100.00** |
| 14 | 98.76 | 93.77 | 98.67 | 97.51 | 98.40 | 97.09 | 99.25 | 95.68 | **99.39** |
| 15 | 81.57 | 47.85 | 73.61 | 72.88 | 79.92 | 79.15 | 74.06 | 84.18 | **90.33** |
| 16 | 77.41 | **98.13** | 90.56 | 88.68 | 91.48 | 94.92 | 96.43 | 95.56 | 95.75 |
| OA | 85.42 ±2.67 | 83.48 ±4.47 | 89.18 ±1.59 | 89.24 ±1.14 | 90.18 ±1.92 | 91.37 ±0.34 | 91.40 ±0.71 | 93.58 ±0.67 | **94.29 ±0.69** |
| AA | 87.01 ±2.58 | 90.82 ±3.47 | 92.80 ±1.85 | 92.98 ±1.09 | 94.09 ±1.31 | 94.88 ±0.76 | 95.37 ±0.54 | 96.11 ±0.76 | **96.92 ±0.98** |
| $\kappa$ | 83.72 ±2.74 | 81.58 ±5.13 | 87.95 ±1.76 | 88.03 ±1.24 | 89.09 ±2.10 | 90.41 ±0.37 | 90.43 ±0.81 | 92.96 ±0.61 | **93.66 ±0.77** |

TABLE V
CLASSIFICATION RESULTS (%) ON HT

| Class | HybridSN [7] | SSRN [8] | DCFSL [22] | Gia-CFSL [18] | RPCL-FSL [19] | CPSRCL [39] | ADAFSL [21] | CDFS-CASCL [34] | QODMA |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 61.97 | 90.84 | 88.24 | 82.22 | **99.04** | 84.35 | 92.34 | 84.95 | 84.03 |
| 2 | 71.49 | 66.71 | 78.94 | 86.59 | 75.64 | 89.67 | 74.66 | 90.65 | **92.63** |
| 3 | 88.27 | 87.51 | 94.87 | 97.83 | 97.91 | 98.77 | 95.95 | 98.86 | **99.57** |
| 4 | 91.03 | 80.02 | 82.49 | 93.38 | 92.56 | **93.99** | 91.28 | 93.30 | 93.14 |
| 5 | 88.74 | 95.34 | 98.79 | 97.82 | 98.59 | 97.53 | 96.93 | 98.63 | **99.60** |
| 6 | 74.93 | 64.07 | 79.06 | 76.09 | 78.82 | 78.44 | 80.31 | **88.75** | 86.09 |
| 7 | 49.87 | 43.78 | 51.49 | 62.43 | 71.64 | 61.68 | 65.95 | 82.15 | **87.29** |
| 8 | **59.78** | 27.13 | 40.92 | 38.54 | 42.34 | 30.71 | 39.63 | 42.45 | 35.55 |
| 9 | 35.82 | 61.17 | 40.86 | 58.42 | 66.95 | 72.33 | 59.98 | 61.79 | **77.23** |
| 10 | 61.39 | 54.67 | 66.98 | 69.60 | 62.51 | 85.23 | 61.54 | 58.22 | **88.38** |
| 11 | 83.33 | 45.01 | 55.77 | 66.34 | **69.58** | 63.29 | 63.29 | 63.46 | 67.40 |
| 12 | 44.80 | 54.13 | 70.81 | 60.67 | **73.31** | 69.22 | 67.92 | 51.38 | 59.77 |
| 13 | 91.86 | 77.29 | 81.36 | 87.39 | 94.52 | **96.66** | 37.92 | 93.43 | 92.03 |
| 14 | 91.87 | 86.45 | 90.19 | 92.55 | 92.33 | 92.67 | 92.55 | **100.00** | **100.00** |
| 15 | 71.76 | 81.14 | 92.14 | 91.47 | 98.43 | 97.94 | 95.11 | **100.00** | **100.00** |
| OA | 65.21 ±3.76 | 66.35 ±2.17 | 71.16 ±3.37 | 74.86 ±1.92 | 78.45 ±1.59 | 78.17 ±1.08 | 73.32 ±0.33 | 77.04 ±0.97 | **81.64 ±0.89** |
| AA | 71.12 ±3.14 | 68.04 ±1.89 | 74.19 ±2.68 | 77.42 ±2.14 | 80.93 ±1.76 | 80.83 ±1.12 | 74.36 ±0.53 | 80.66 ±0.65 | **84.18 ±0.66** |
| $\kappa$ | 62.38 ±3.02 | 63.75 ±2.35 | 68.87 ±3.61 | 72.87 ±2.06 | 76.72 ±1.64 | 76.44 ±1.15 | 71.12 ±0.37 | 75.23 ±1.03 | **80.19 ±0.96** |

the 16 ground-truth classes, our QODMA achieves the best result in nine classes. Our method achieved better classification performance on Classes 10 "Corn_senesced_green_weeds" and 14 "Lettuce_romaine_7wk," which are spatially adjacent and exhibit similar spectral properties.

Table V presents the classification results of various methods on the HT dataset. Our proposed QODMA method also provides the best results. Compared with CDFS-CASCL, QODMA improves the OA, AA, and $\kappa$ by 4.6%, 3.52%, and 4.96%, respectively. While both CDFS-CASCL and QODMA incorporate textual information, the key difference is that QODMA addresses the inconsistency in multimodal feature spaces and the issue of domain shift. Additionally, the improved metric approach refines the feature space and enhances the final classification results.

Figs. 4–7 illustrate the classification maps generated by various methods. It is obvious that our approach yields results more consistent with the ground truth. In particular, we used red rectangles to highlight the regions where our QODMA significantly outperforms other methods.

To evaluate the effectiveness of our proposed QODMA method with different numbers of labeled samples, we select one–five labeled samples in each class of the target domain for training and the rest for testing. Fig. 8 shows the OA

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TABLE VI
MCNEMAR'S TEST ON FOUR DATASETS TO VALIDATE WHETHER THE DIFFERENCE BETWEEN QODMA AND EACH OF OTHER METHODS IS STATISTICALLY SIGNIFICANT

| | | HybridSN | SSRN | DCFSL | Gia-CFSL | RPCL-FSL | CPSRCL | ADAFSL | CDFS-CASCL |
|---|---|---|---|---|---|---|---|---|---|
| IP | OA | 64.22 | 66.16 | 63.77 | 66.25 | 75.67 | 74.45 | 70.55 | 80.67 |
| | Z | -28.33 | -26.44 | -29.75 | -26.27 | -13.72 | -15.42 | -21.02 | -5.68 |
| UP | OA | 72.96 | 76.78 | 78.66 | 80.41 | 83.86 | 83.69 | 81.62 | 81.81 |
| | Z | -57.26 | -44.87 | -39.09 | -33.26 | -20.16 | -21.32 | -28.77 | -27.85 |
| SA | OA | 85.42 | 83.48 | 89.18 | 89.24 | 90.18 | 91.37 | 91.40 | 93.58 |
| | Z | -47.16 | -58.23 | -30.81 | -30.22 | -25.63 | -19.14 | -18.95 | -7.79 |
| HT | OA | 65.21 | 66.35 | 71.16 | 74.86 | 78.45 | 78.17 | 73.32 | 77.04 |
| | Z | -29.15 | -26.97 | -20.54 | -13.26 | -5.74 | -6.41 | -15.79 | -9.62 |

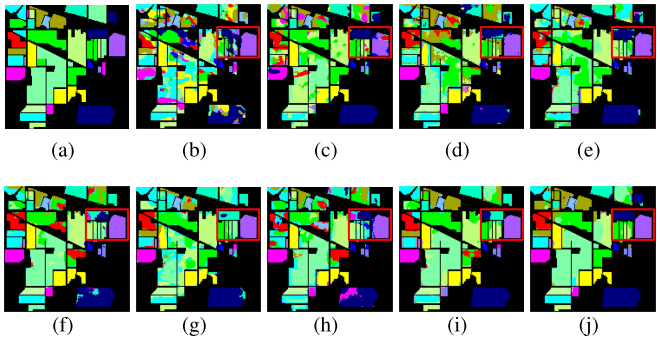


Fig. 4. Classification map of different methods on IP. (a) Ground-truth, (b) HybridSN, (c) SSRN, (d) DCFSL, (e) Gia-CFSL, (f) RPCL-FSL, (g) CPSRCL, (h) ADAFSL, (i) CDFS-CASCL, and (j) QODMA.

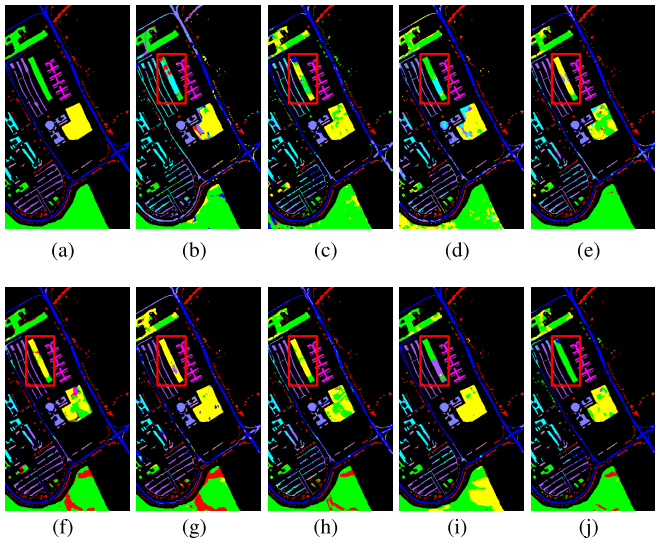


Fig. 5. Classification map of different methods on UP. (a) Ground-truth, (b) HybridSN, (c) SSRN, (d) DCFSL, (e) Gia-CFSL, (f) RPCL-FSL, (g) CPSRCL, (h) ADAFSL, (i) CDFS-CASCL, and (j) QODMA.

of all FSL-based methods on IP, UP, SA, and HT datasets with different numbers of labeled samples per class. It can be observed that the OA increases with the number of labeled samples, and our QODMA method consistently outperforms other approaches.

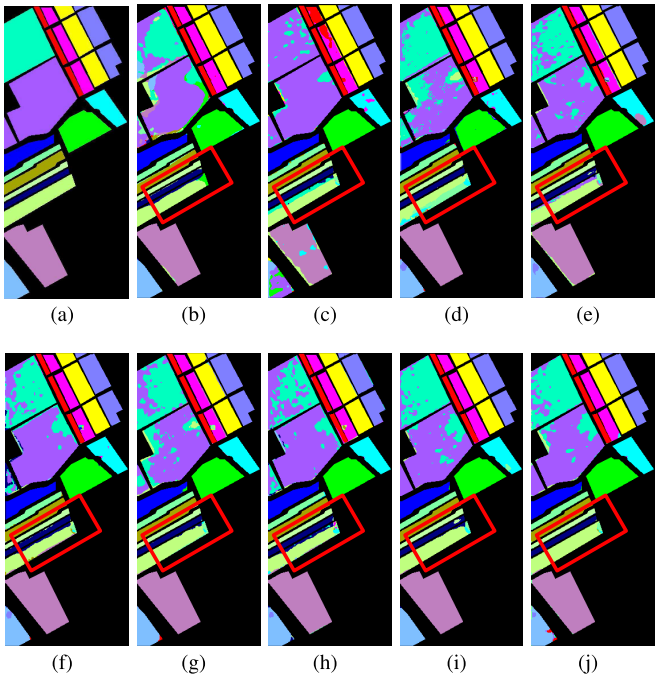


Fig. 6. Classification map of different methods on SA. (a) Ground-truth, (b) HybridSN, (c) SSRN, (d) DCFSL, (e) Gia-CFSL, (f) RPCL-FSL, (g) CPSRCL, (h) ADAFSL, (i) CDFS-CASCL, and (j) QODMA.

### D. Statistical Analysis

To assess whether the performance differences between QODMA and other methods are statistically significant, we adopt McNemar's test [40]. The test evaluates the $Z$-statistic based on the number of samples misclassified by one method but correctly classified by the other. At the 5% significance level, $|Z| > 1.96$ indicates a significant difference, with negative values favoring QODMA. As shown in Table VI, all datasets yield $Z < -1.96$, confirming that QODMA achieves statistically significant improvements.

### E. Ablation Analysis

The proposed QODMA method uses the QODA module to maintain multimodal space consistency, the DML module

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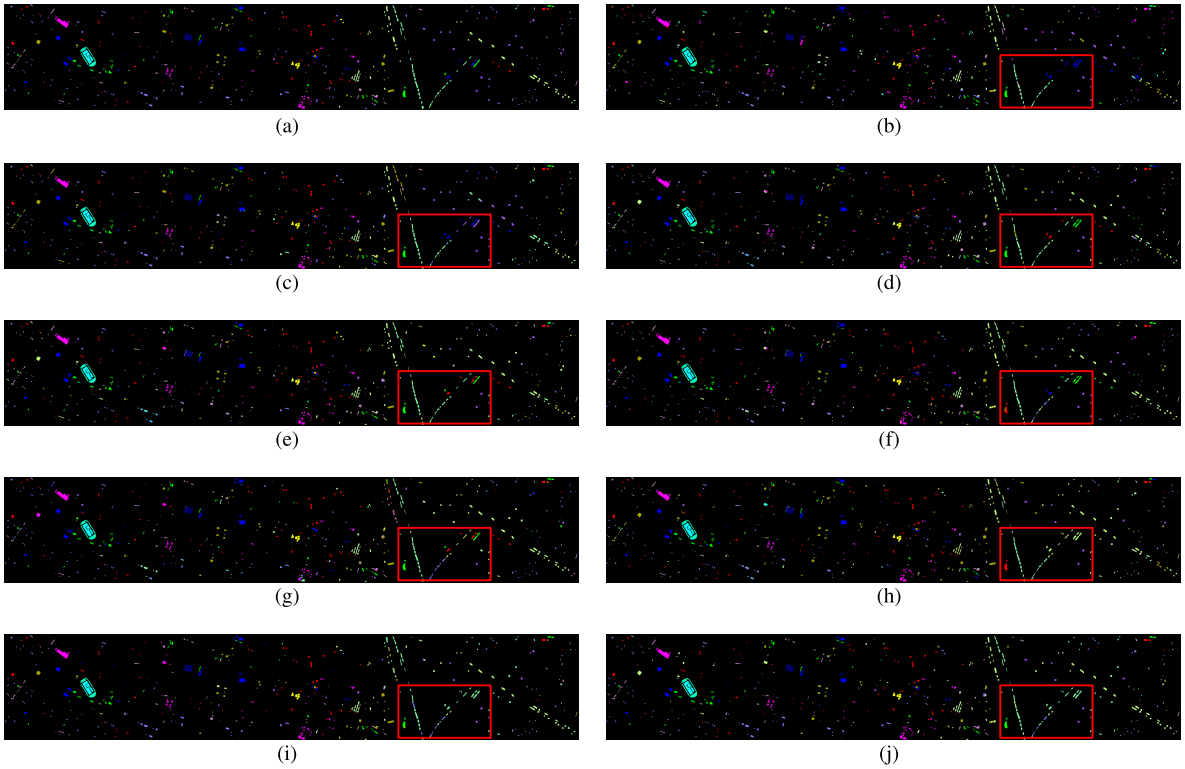



Fig. 7. Classification map of different methods on HT. (a) Ground-truth, (b) HybridSN, (c) SSRN, (d) DCFSL, (e) Gia-CFSL, (f) RPCL-FSL, (g) CPSRCL, (h) ADAFSL, (i) CDFS-CASCL, and (j) QODMA.

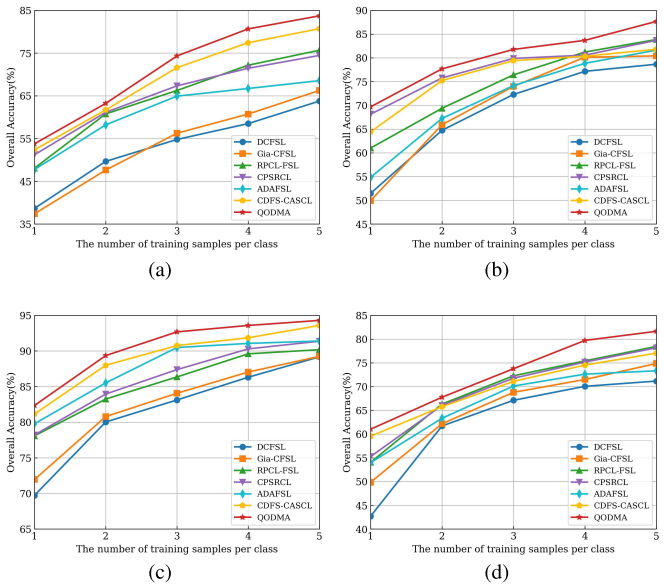



Fig. 8. OA (%) of all methods under different numbers of labeled samples per class in the target domain dataset. (a) IP, (b) UP, (c) SA, and (d) HT.

to capture feature dependencies, and the IDSCL module to alleviate domain shift. To evaluate the performance of each module, we conduct ablation experiments on four datasets, and the results are presented in Table VII.

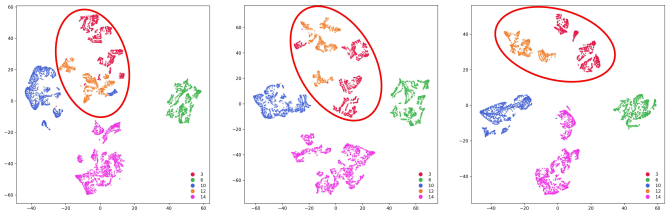
It can be observed that the EU and BDC are complementary, and the DML module, by combining these two metrics, improves the original EU-based method by 2.51%, 2.21%, 1.19%, and 2.34% across four datasets, respectively. By further adding QODA or IDSCL into the DML, the OA further improves. In particular, the QODA module plays a dominant role, especially for the IP dataset. In general, each module contributes to the FSL framework, and the integration of all modules yields the best overall performance.

TABLE VII
ABLATION ANALYSIS OF DIFFERENT MODULES OF QODMA

| | DML | | QODA | IDSCL | OA(%) | | | |
|---|---|---|---|---|---|---|---|---|
| | EU | BDC | | | IP | UP | SA | HT |
| Methods | ✓ | | | | 72.91 | 82.32 | 91.27 | 76.83 |
| | | ✓ | | | 73.28 | 83.07 | 91.94 | 77.96 |
| | ✓ | ✓ | | | 75.42 | 84.53 | 92.46 | 79.17 |
| | ✓ | ✓ | | ✓ | 77.16 | 85.79 | 93.07 | 80.27 |
| | ✓ | ✓ | ✓ | | 82.78 | 86.83 | 93.74 | 81.08 |
| | ✓ | ✓ | ✓ | ✓ | 83.72 | 87.68 | 94.29 | 81.64 |

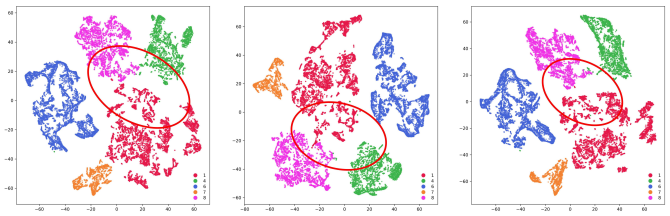
To further visualize the effectiveness of IDSCL and QODA modules, we use the T-distributed stochastic neighbor embedding (t-SNE) to display the features. Figs. 9(a)–(c)–12(a)–(c) are the t-SNE features of different combinations of modules. It can be seen that both IDSCL and QODA modules improve the discriminate ability, and the QODA module generates more compact intraclass features and more dispersed interclass features.

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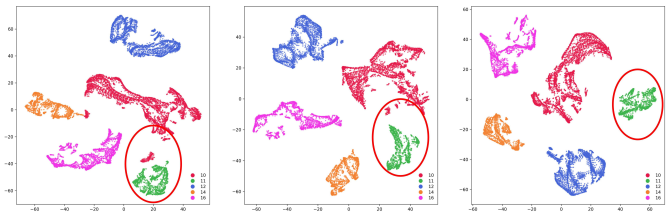
(a) (b) (c)

Fig. 9. Feature visualization of different modules using the t-SNE on IP. (a) Basic FSL framework, (b) adding the IDSCL module to the basic FSL framework, and (c) adding the QODA module to the basic FSL framework.



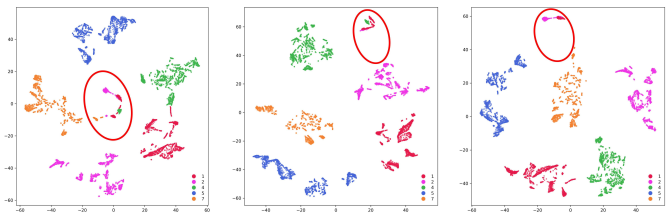
(a) (b) (c)

Fig. 10. Feature visualization of different modules using the t-SNE on UP. (a) Basic FSL framework, (b) adding the IDSCL module to the basic FSL framework, and (c) adding the QODA module to the basic FSL framework.



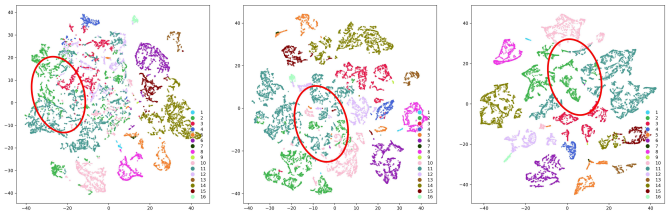
(a) (b) (c)

Fig. 11. Feature visualization of different modules using the t-SNE on SA. (a) Basic FSL framework, (b) adding the IDSCL module to the basic FSL framework, and (c) adding the QODA module to the basic FSL framework.



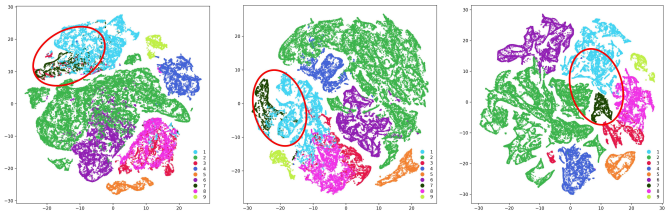
(a) (b) (c)

Fig. 12. Feature visualization of different modules using the t-SNE on HT. (a) Basic FSL framework, (b) adding the IDSCL module to the basic FSL framework, and (c) adding the QODA module to the basic FSL framework.



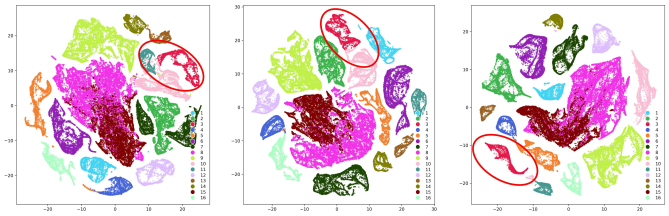
(a) (b) (c)

Fig. 13. Feature visualization of different methods using the t-SNE on IP. (a) Gia-CFSL, (b) CPSRCL, and (c) QODMA.



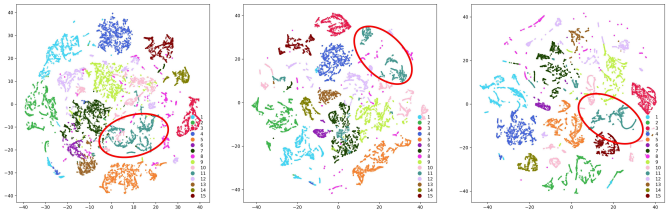
(a) (b) (c)

Fig. 14. Feature visualization of different methods using the t-SNE on UP. (a) Gia-CFSL, (b) CPSRCL, and (c) QODMA.



(a) (b) (c)

Fig. 15. Feature visualization of different methods using the t-SNE on SA. (a) Gia-CFSL, (b) CPSRCL, and (c) QODMA.



(a) (b) (c)

Fig. 16. Feature visualization of different methods using the t-SNE on HT. (a) Gia-CFSL, (b) CPSRCL, and (c) QODMA.

### *F. Feature Visualization*

Here, we use the t-SNE figures to visually compare some FSL-based methods (Gia-CFSL, CPSRCL, and QODMA) on four datasets in Figs. 13–16, respectively. Compared with Gia-CFSL and CPSRCL methods, the proposed QODMA achieves more compact intraclass and more dispersed inter-class features, with less feature confusion across the four datasets. For example, in Fig. 13 for the IP dataset, it is difficult to distinguish the Classes 11 "Soybean-mintill" and 12 "Soybean-clean" due to their similar features. Our proposed QODMA method effectively distinguishes these two classes, as shown in the red circled region. Similarly, on the UP dataset, Classes 1 "Asphalt" and 7 "Bitumen" are difficult to be distinguished. From Fig. 14(a)–(c), our method shows clear feature distribution boundaries between these two classes. On the SA and HT datasets, our method also has large inter-class scatter. The aforementioned results clearly indicate that our proposed QODMA method effectively reduces intraclass variations while increasing inter-class separability.

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TABLE VIII
COMPUTATIONAL TIME (S), FLOPS (M), AND PARAMETERS (M) OF DIFFERENT METHODS ON THREE DATASETS

| Dataset | Metric | Methods | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | DCFSL | Gia-CFSL | RPCL-FSL | CPSRCL | ADAFSL | CDFS-CASCL | QODMA |
| IP | Training time | 1928.31 | 4492.75 | 736.89 | 1128.54 | 3718.63 | 2893.49 | 3327.61 |
| | Testing time | 1.89 | 1.93 | 3.59 | 3.93 | 1.99 | 1.47 | 2.13 |
| | FLOPS | 48.54 | 26837.44 | 48.53 | 786.74 | 43.84 | 257.64 | 373.09 |
| | #Params | 4.26 | 0.85 | 4.27 | 4.28 | 4.25 | 0.93 | 2.02 |
| UP | Training time | 1378.54 | 2711.06 | 629.94 | 959.91 | 2605.53 | 1831.53 | 2442.21 |
| | Testing time | 10.74 | 7.19 | 16.16 | 16.88 | 7.46 | 5.49 | 9.64 |
| | FLOPS | 47.72 | 26821.54 | 47.72 | 783.54 | 43.02 | 256.89 | 372.14 |
| | #Params | 4.25 | 0.84 | 4.26 | 4.26 | 4.24 | 0.91 | 2.01 |
| SA | Training time | 1951 | 4591.48 | 998.24 | 1428.24 | 3889.27 | 2912.96 | 3591.12 |
| | Testing time | 13.36 | 10.57 | 22.03 | 24.11 | 12.25 | 8.17 | 16.23 |
| | FLOPS | 48.54 | 26839.67 | 48.54 | 788.31 | 43.84 | 258.37 | 375.17 |
| | #Params | 4.26 | 0.85 | 4.27 | 4.28 | 4.25 | 0.93 | 2.01 |
| HT | Training time | 1755.13 | 4103.27 | 616.48 | 1064.13 | 3184.31 | 2659.28 | 3027.23 |
| | Testing time | 2.53 | 2.56 | 5.48 | 5.49 | 2.67 | 2.94 | 4.43 |
| | FLOPS | 47.81 | 26831.39 | 47.96 | 781.26 | 43.53 | 257.13 | 372.17 |
| | #Params | 4.25 | 0.84 | 4.26 | 4.27 | 4.24 | 0.91 | 2.01 |

### G. Analysis of Computational Complexity

Table VIII lists the training time, testing time, floating point operations (FLOPs), and the number of parameters of different FSL methods. Due to the additional metric learning processes and cross-modal feature interactions, our QODMA method needs more training times. However, the training time of our method is still shorter than some classical FSL methods, i.e., Gia-CFSL and ADAFSL. The testing time, FLOPs, and the number of parameters of different methods are comparable. Overall, our QODMA achieves the best classification performance while maintaining an acceptable computational cost.

## IV. CONCLUSION

In this article, we have proposed a QODMA method for few-shot HSIC. It designs a QODA module to mitigate semantic discrepancies between visual and textual features using bidirectional attention and contrastive learning. To capture both local similarities and higher-order dependencies, a DML module is further constructed to combine the ED with the BDC to enhance the model's discriminative ability. Furthermore, an IDSCL is designed to align features between the source and target domains. Experimental results show that QODMA outperforms state-of-the-art few-shot methods for HSIC. While QODMA demonstrates significant advancements in few-shot HSIC, it faces two main challenges. First, the dual-branch design and covariance-based metric introduce extra computational overhead. Second, IDSCL focuses on second-order correlations and may overlook higher-order feature interactions. Future work will address these issues by exploring efficient covariance approximations and tensor-based modeling of higher-order relations.

## REFERENCES

[1] A. Plaza et al., "Recent advances in techniques for hyperspectral image processing," *Remote Sens. Environ.*, vol. 113, pp. S110–S122, 2009.

[2] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[3] J. Peng, W. Sun, and Q. Du, "Self-paced joint sparse representation for the classification of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1183–1194, Feb. 2019.

[4] J. Peng, L. Li, and Y. Y. Tang, "Maximum likelihood estimation-based joint sparse representation for the classification of hyperspectral remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 6, pp. 1790–1802, Jun. 2019.

[5] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[6] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[7] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[8] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[9] S. Mei, J. Ji, J. Hou, X. Li, and Q. Du, "Learning sensor-specific spatial–spectral features of hyperspectral images via convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4520–4533, Aug. 2017.

[10] X. Yang, Y. Ye, X. Li, R. Y. K. Lau, X. Zhang, and X. Huang, "Hyperspectral image classification with deep learning models," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5408–5423, Sep. 2018.

[11] Q. Liu, J. Peng, G. Zhang, W. Sun, and Q. Du, "Deep contrastive learning network for small-sample hyperspectral image classification," *J. Remote Sens.*, vol. 3, p. 0025, Jan. 2023.

[12] O. Vinyals, C. Blundell, T. Lillicrap, K. Kavukcuoglu, and D. Wierstra, "Matching networks for one shot learning," in *Proc. NIPS*, vol. 29, 2016, pp. 3637–3645.

[13] Q. Sun, Y. Liu, T.-S. Chua, and B. Schiele, "Meta-transfer learning for few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 403–412.

[14] J. Sun, Y. Sun, X. Shen, and Q. Sun, "Query-guided prototype optimization for few-shot classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 12, pp. 12197–12209, Dec. 2024.

[15] M. Huisman, J. N. van Rijn, and A. Plaat, "A survey of deep meta-learning," *Artif. Intell. Rev.*, vol. 54, no. 6, pp. 4483–4541, Aug. 2021.

[16] J. Snell, K. Swersky, and R. S. Zemel, "Prototypical networks for few-shot learning," in *Proc. NIPS*, vol. 30, 2017, pp. 4080–4090.

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[17] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2019.

[18] Y. Zhang, W. Li, M. Zhang, S. Wang, R. Tao, and Q. Du, "Graph information aggregation cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 1912–1925, Feb. 2024.

[19] Q. Liu et al., "Refined prototypical contrastive learning for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506214.

[20] Z. Li et al., "Few-shot hyperspectral image classification with self-supervised learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5517917.

[21] Z. Ye, J. Wang, H. Liu, Y. Zhang, and W. Li, "Adaptive domain-adversarial few-shot learning for cross-domain hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5532017.

[22] Z. Li et al., "Deep cross-domain few-shot learning forhyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5501618.

[23] J. Li, Z. Zhang, R. Song, Y. Li, and Q. Du, "SCFormer: Spectral coordinate transformer for cross-domain few-shot hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 33, pp. 840–855, 2024.

[24] Q. Zhu, H. Li, W. Deng, Q. Guan, and J. Luo, "From intra-distinctiveness to inter-invariance: A cycle-resemblance few-shot transformation network for cross-domain hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5507616.

[25] D. Wertheimer and B. Hariharan, "Few-shot learning with localization in realistic settings," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6558–6567.

[26] W. Li, L. Wang, J. Huo, Y. Shi, Y. Gao, and J. Luo, "Asymmetric distribution measure for few-shot learning," in *Proc. IJCAI*, 2020, pp. 2957–2963.

[27] B. Xi, J. Li, Y. Li, R. Song, D. Hong, and J. Chanussot, "Few-shot learning with class-covariance metric for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 5079–5092, 2022.

[28] J. Sun, X. Shen, and Q. Sun, "Hyperspectral image few-shot classification network based on the Earth mover's distance," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5534114.

[29] X. Chen, N. Rostamzadeh, B. N. Oreshkin, and P. O. Pinheiro, "Adaptive cross-modal few-shot learning," in *Proc. NIPS*, vol. 32, 2019, pp. 4847–4857.

[30] M. Afham, S. Khan, M. Haris Khan, M. Naseer, and F. Shahbaz Khan, "Rich semantics improve few-shot learning," 2021, *arXiv:2104.12709*.

[31] K. Nishida, K. Nishida, and S. Nishioka, "Improving few-shot image classification using machine-and user-generated natural language descriptions," 2022, *arXiv:2207.03133*.

[32] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. PMLR*, 2021, pp. 8748–8763.

[33] Z. Li, H. Tang, Z. Peng, G.-J. Qi, and J. Tang, "Knowledge-guided semantic transfer network for few-shot image recognition," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Feb. 6, 2023, doi: 10.1109/TNNLS.2023.3240195.

[34] Z. Li et al., "Cross-domain few-shot hyperspectral image classification with cross-modal alignment and supervised contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5519319.

[35] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.

[36] G. J. Székely, M. L. Rizzo, and N. K. Bakirov, "Measuring and testing dependence by correlation of distances," *Ann. Statist.*, vol. 35, no. 6, pp. 2769–2794, Dec. 2007.

[37] G. J. Székely and M. L. Rizzo, "Brownian distance covariance," *Ann. Appl. Statist.*, vol. 3, no. 4, pp. 1236–1265, Dec. 2009.

[38] J. Xie, F. Long, J. Lv, Q. Wang, and P. Li, "Joint distribution matters: Deep Brownian distance covariance for few-shot classification," in *Proc. CVPR*, Jun. 2022, pp. 7962–7971.

[39] Q. Liu, J. Peng, N. Chen, W. Sun, Y. Ning, and Q. Du, "Category-specific prototype self-refinement contrastive learning for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5524416.

[40] G. Moody, "Thematic map comparison: Evaluating the statistical significance of differences in classification accuracy," *Photogramm. Eng. Remote Sens.*, vol. 70, no. 5, pp. 627–633, 2004.

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